SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CONCEPT MAPPING

Course Code:	CSA1613	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	**Assessment:**	Assessment Tool 3 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (5 Questions × 100 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	M.Dilip Kumar	**Reg. No.:**	192525038
Section / Batch:		**Date:**	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled..

Question Type	Focus Area	Questions	Marks
Concept Mapping	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2,Q3,Q4,Q5	100
		GRAND TOTAL:	**100 Marks**

SECTION A – CONCEPT MAPPING

Concept Map 1: Decision Support Systems (DSS) Task

Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decisionmaking.
Core components: Data, Model, User Interface

1. Types of DSS
2. Flow: Data → Analysis → Decision
3. Role in business decision-making
4. Real-world example linkage

Concept Map 2: Operational Systems vs DSS Task

Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse Task

Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME) Task

Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration

Code of Conduct
I am M.Dilip Kumar(192525038)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: __dilip______________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

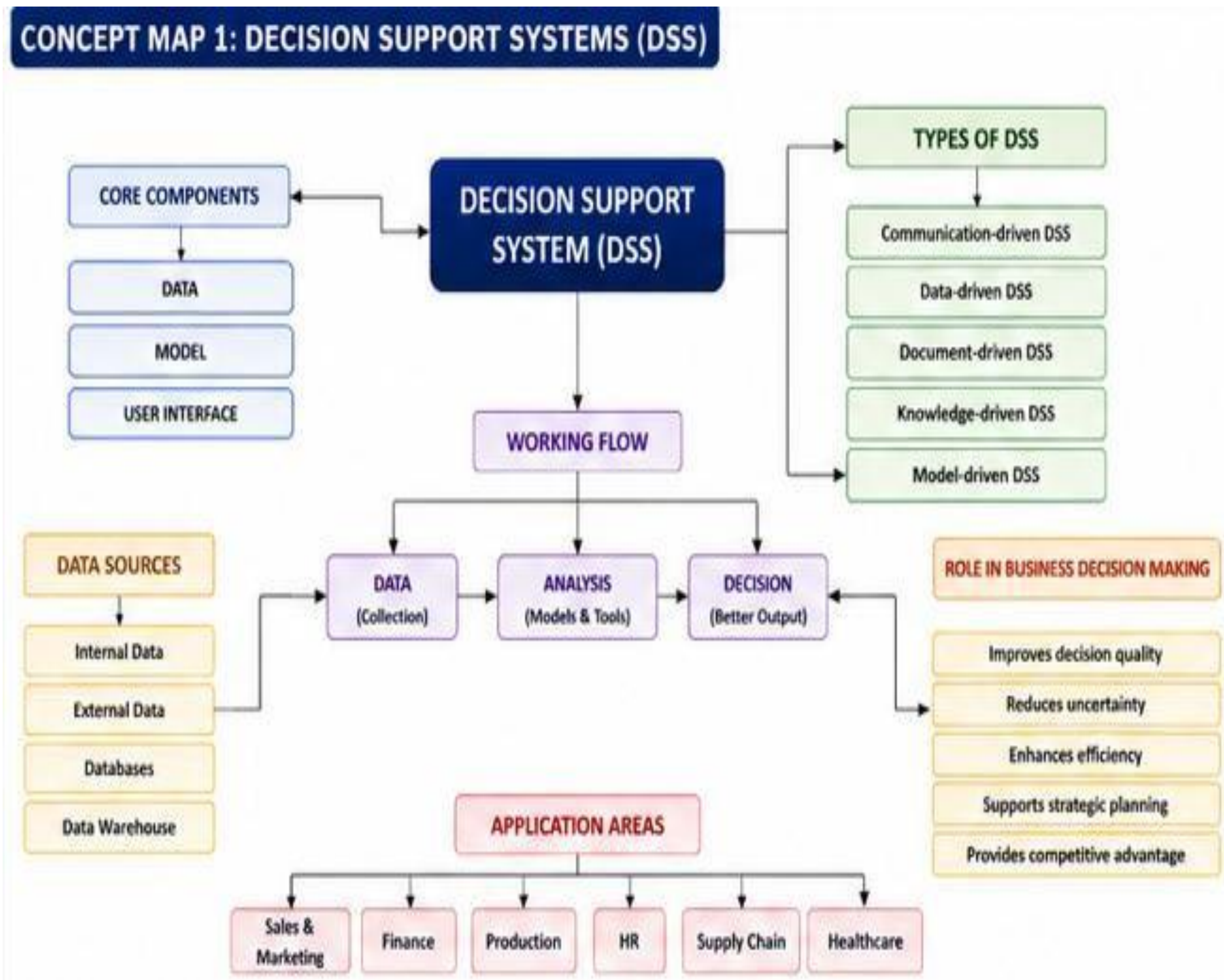
Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Q3						
Q4						
Q5						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Concept Map 1: Decision Support Systems (DSS) Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decisionmaking.

Core components: Data, Model, User Interface



Concept Map 2: Operational Systems vs DSS Task Description:

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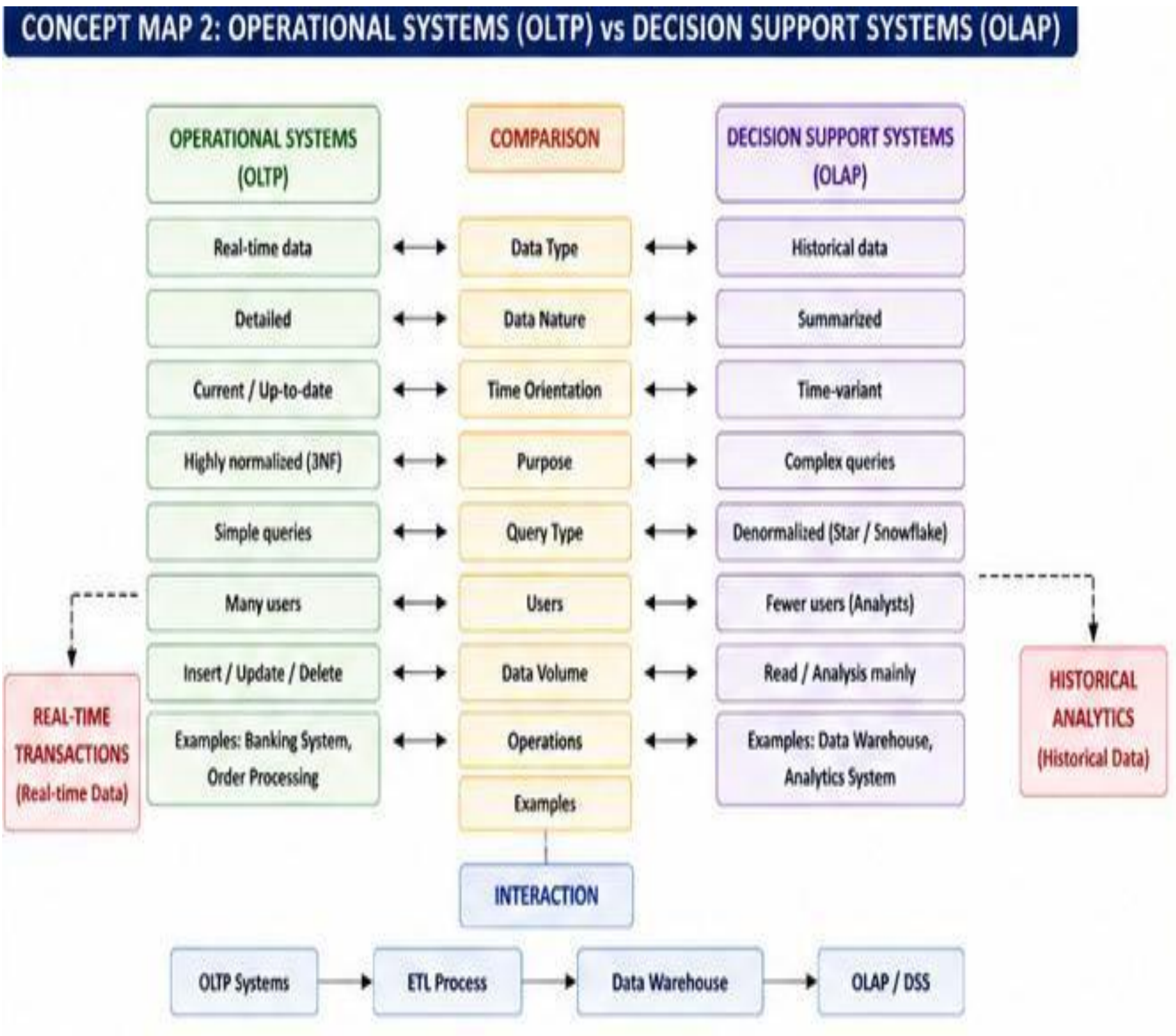
4.



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Data Sources → ETL → Data Warehouse → Data Marts

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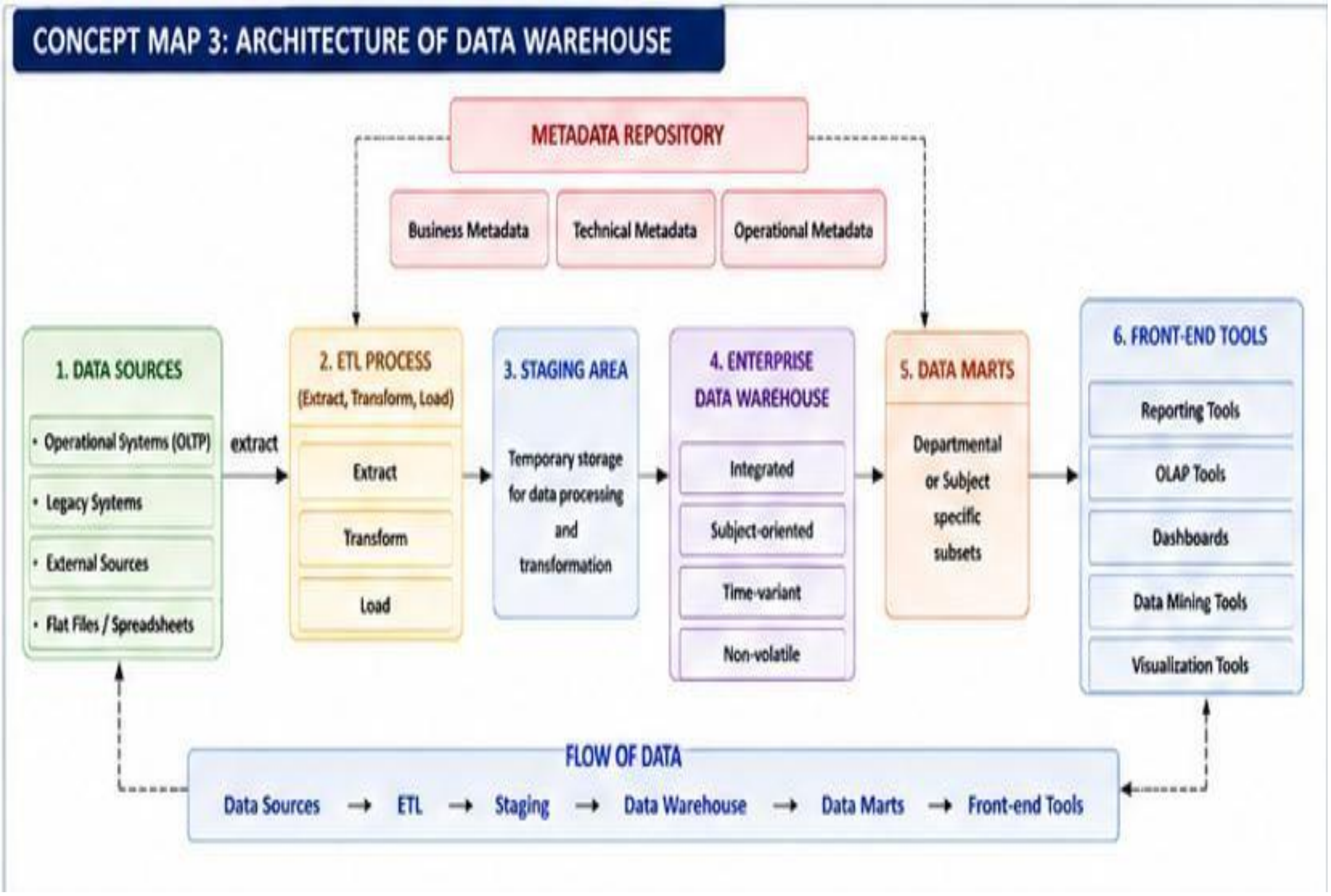

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Data cube structure (dimensions, measures)

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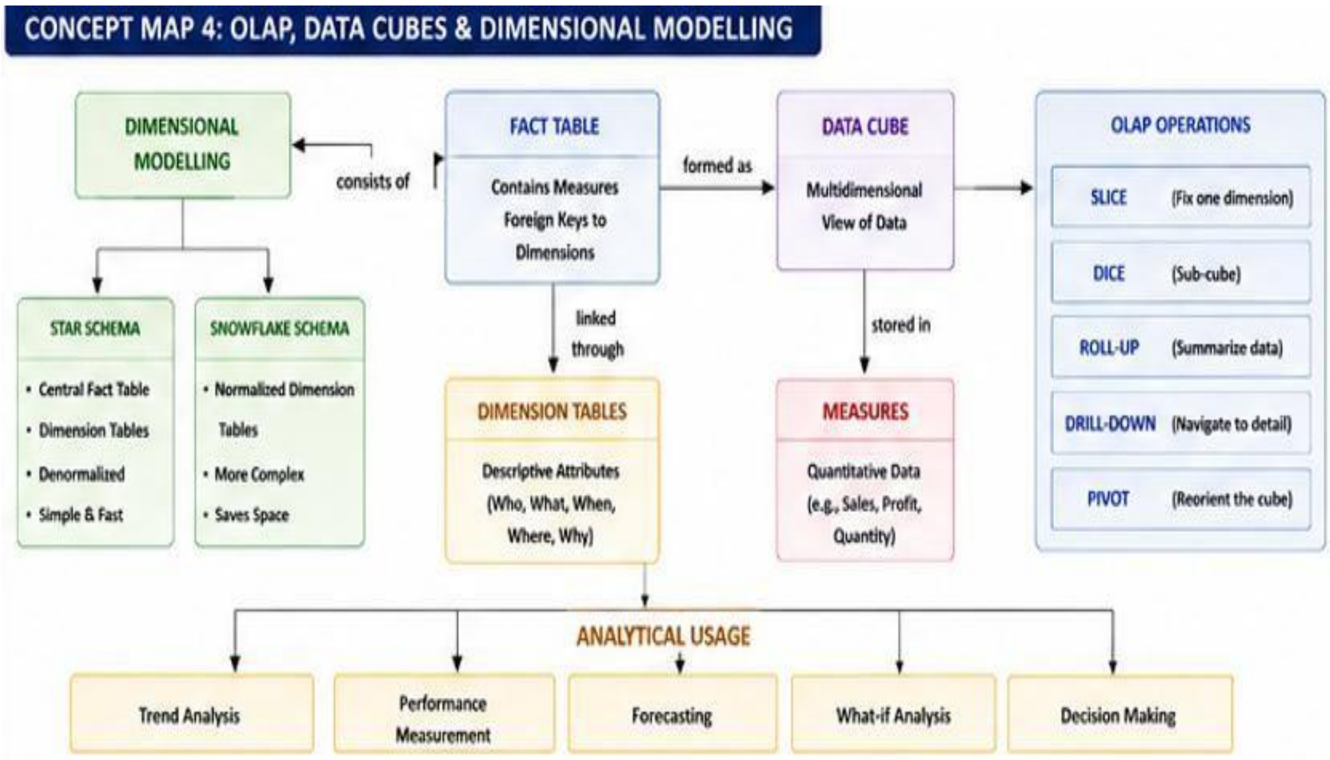
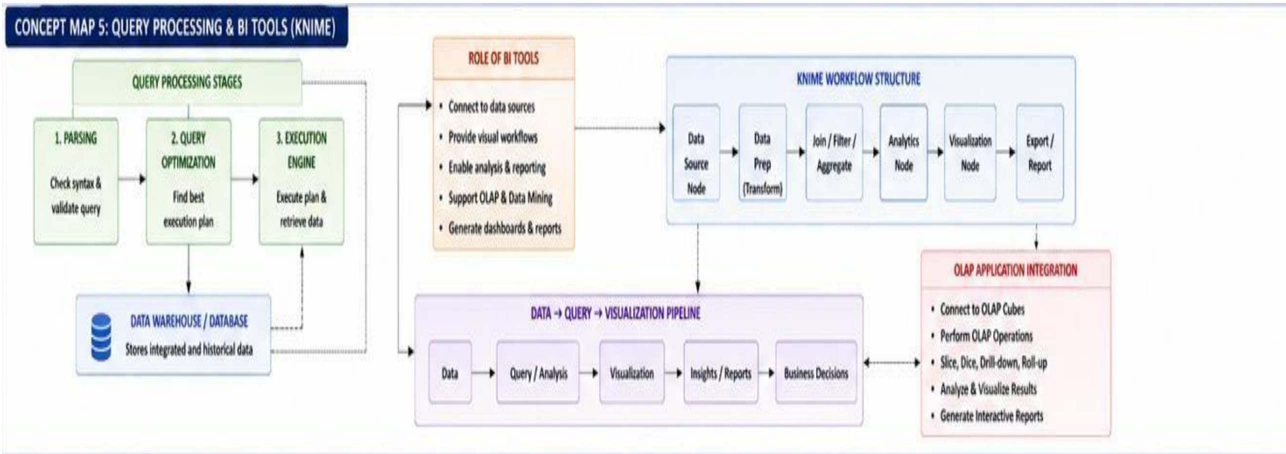


Concept Map 5: Query Processing & BI Tools (KNIME) Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

Query processing stages (Parsing, Optimization, Execution)

1. Role of BI tools
2. KNIME workflow structure
3. Data → Query → Visualization pipeline
4. OLAP application integration


CONCEPT MAP 5: QUERY PROCESSING & BI TOOLS (KNIME)
QUERY PROCESSING STAGES
1. PARSING
Check syntax & validate query
2. QUERY OPTIMIZATION
Find best execution plan
3. EXECUTION ENGINE
Execute plan & retrieve data
DATA WAREHOUSE / DATABASE
Stores integrated and historical data
ROLE OF BI TOOLS
Connect to data sources
Provide visual workflows
Enable analysis & reporting
Support OLAP & Data Mining
Generate dashboards & reports
KNIME WORKFLOW STRUCTURE
Data Source Node
Data Prep (Transform)
Join / Filter / Aggregate
Analytics Node
Visualization Node
Export / Report
DATA → QUERY → VISUALIZATION PIPELINE
Data
Query / Analysis
Visualization
Insights / Reports
Business Decisions
OLAP APPLICATION INTEGRATION
Connect to OLAP Cubes
Perform OLAP Operations
Slice, Dice, Drill-down, Roll-up
Analyze & Visualize Results
Generate Interactive Reports